



BRAINWARE UNIVERSITY

398, Ramkrishnapur Road, Barasat, North 24 Parganas, Kolkata - 700 125

Assignment Submission

Assignment No: - 01

Name of the Department:- Computer Science and Engineering

Programme Name: - BTech CSE

Semester / year:- 3rd Semester

Course Code: - BTS30102

Course Name: - Database Management Systems

Answer the following Questions:-

			Marks
Question no 1	:	A company is developing an Integrated E-Commerce and Healthcare Platform to manage customers, medicines, pharmacies, hospitals, doctors, patients, appointments, prescriptions, payments, and insurance claims. As a Database Designer, perform the following: a) Develop an EER diagram containing at least 10 entities. Include appropriate relationships, cardinalities, a weak entity, ISA relationship, and composite, multivalued, and derived attributes. b) Convert the EER model into a relational database schema and identify primary keys, candidate keys, composite keys, foreign keys, and integrity constraints. c) Evaluate the functional dependencies and identify partial and transitive dependencies. Determine candidate keys where applicable. b) Normalize the selected relations from 1NF to 3NF and explain how normalization eliminates insertion, deletion, and update anomalies.	[3]
Question no 2	:	Develop relational algebra expressions for the following queries: (Using the database designed in Question 1), - Find high-value customers. - Find delayed orders. - Find medicines with low or unavailable stock. - Find rejected insurance claims. - Find doctors with a high number of patient visits. - Find pharmacies supplying a particular medicine.	[2]

Name of the Subject Teacher: Mr. Ataul Hoque Mandal

Designation: Assistant Professor

Date of notification: 19/08/2026

● **Last date of Submission: - 28/08/2026**